Complete Empirical Capture of the Agent/Environment Interaction during Natural Behavior across Phylogony, Domain, and Dimension

FreeMoCap Foundation, Inc.

Topic: Scientific Instrumentation for Sensing and Imaging

1. Mission

- **Funnel**: The modern landscape of science is one of loosely connected pockets of hyper specialization within siloed discplines and domains of inquiry.
- - **Gap**: While this approach has led to many advances over the past century, the current model of specialized single-domain advancemnt has failed to create integrative frameworks to combine the individual threads of single-modality advancement into an coherent whole representing a richly interconnected representation nature capable of actionable operation.
- **Hero**: A high quality integrated platform directly designed for the use-case development of complete empirical capture of an awake behaving human-or-non-human-animal (HNHA) would transform the scientific landscape by creating a technical commons whereby specialists and cross-disciplnary researchers can productively interact in serivce of the shared and unified scienfitic envdeavor.

MISSION STATEMENT - Build a Platform for **Complete Empirical Capture of the Agent/Environment Interaction** to create Interoperabilty across all Dimensional Domains to align Neuroscience with Robotics

Complete Empirical Capture: Complete empirical capture means that we are recording every single possible thing that we can record about a given slice of time over continuous time.
Agent/Environment Interaction: Relates to the internal and external state of an actor (human, non-human animal, or artificial system).
Interoperabilty across all Dimensional Domains: A "dimensional domain" is any thing that can vary along an orientable dimension. We will a platform that will create seamless interoperability across **species**, **scale**, **time**, **numerosity**, and **complexity**.

- Estimate per timestep-
  - State
  - Inputs
  - Outputs
  - Environment

2. Technology Landscape

- Existing tools and methods
  - Myriad single-modality hyperspecialized research threads
    - OpenSim - Widely used but universally hated by practictioners
    - DeepLabCut - Global success on paper, universally a headache in practice
    - etc
- New approaches to complexity management
  - Ontology/Entity Management (Anduril/Palantir/Maven Smart System)
  - Industry and Military technology applied to scientific domain
  - Want to establish an operational ontology as backbone of this tool, following powerful patterns established by (Anduril/Palantir/Maven) systems
- FreeMoCap and its landscape
  - Established Tool with Growing International Community
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
- Laser Skeleton research
  - (Matthis et al., 2018; 2022; Muller et al., 2023)
  - etc
  - Ferret work
    -

3. Outcomes

- Cool Tool - ~~Skinner~~SkellyBox: Multiscale modular empirical capture arena
  - Interoperabilty across species, domains, and timescales
    - A member of a mouse lab can walk into a marmoset lab and expect their pipelines will still work
    - Perceptuomotor behavior and kinematic/kinetic analysis directly interoperable with robotics models (via MuJoCo and Nvidia Isaac Gym) - Data for reinforcement learning models, Controls to interpret HNHA behavioral data
    - Data models alignment across timescales - developmental/longitudinal (months/years/lifespan) timescales aligned with perception/action timescales (seconds/minutes)
  - Composable complexity
    - Careful ontology allows for complexity via comopsible data sources and assocaited downstream entities
    - Adding new sensor modalities into established scene, framework handles spatiotemporal alignment/calibration, unit conversions, resampling, etc
- Organize and operationalize a new culture of science around discussion/management/extension shared platform tool
  - In the form of an annual Congress meeting
  - Organizing the culture of science around the discussion and advancement of shared ontologies rather than an endless stream of 8-10 page pdfs
  - Break the Skill Ceiling in academic science
    - Build tool to allow multi-decade continuous education across disciplines and experience (same tool in high school as used in federally funded research lab)
    - Shared toolset encourages natural blending of disciplines while encouraging and reward higher specialized exploration (esp when that research produces actionable results across different dimensional domains)
- Mesoscale revolution
  - Allow advances in reductionist microscale research to align with human-scale interests
  - Align Perceptuomotor Neuroscience with humanoid robotics, allow transfer, technology, and technique transfer between those fields

3.1. 7 Year Plan

1. Organize - Set up Organization (Phase 0) - Solidify release of FreeMoCap v2
2. Architect - Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research) - Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)
3. Execute - Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3) - Iterative design of instrumented system
4. Stabilize -
5. Expand
6. Align

7. Establish
8. Maintain

4. Senior/Key Personnel Qualifications

Provide a brief description of the qualifications of Senior/Key Personnel on the team. Include technical expertise, experience commercializing technology, past efforts, and other qualifications that will contribute to achieving the team's Mission. Include information for all Senior/Key Personnel, including team members not currently affiliated with the lead organization.

5. Team Capabilities Statement

Provide a Team Capability Statement describing how the Senior/Key Personnel on the proposed leadership team bring together complementary expertise in strategic leadership, technical expertise, and/or entrepreneurship, as applicable. The statement should describe how, collectively, the team demonstrates the experience, adaptability, and collaborative approach necessary to execute the Mission. Briefly describe the team's intended governance structure and autonomy during Phase 0 and Phase 1. If applicable, include an overview of networks, partnerships, and capital resources your team currently has and how they will be deployed in support of the proposal.

Bibliography

Matthis, J. S., Muller, K. S., Bonnen, K. L., & Hayhoe, M. M. (2022). Retinal Optic Flow during Natural Locomotion. *PLOS Computational Biology*, *18*(2), e1009575. https://doi.org/10.1371/journal.pcbi.1009575

Matthis, J. S., Yates, J. L., & Hayhoe, M. M. (2018). Gaze and the Control of Foot Placement When Walking in Natural Terrain. *Current Biology*, *28*(8), 1224–1233. https://doi.org/10.1016/j.cub.2018.03.008

Muller, K. S., Matthis, J., Bonnen, K., Cormack, L. K., Huk, A. C., & Hayhoe, M. (2023). Retinal Motion Statistics during Natural Locomotion. *Elife*, *12*, e82410. https://doi.org/10.7554/eLife.82410